

Vocabulary Knowledge Mediates the Link between Socioeconomic Status and Word Learning in  
Grade School.

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### Abstract

The relationship between children's slow vocabulary growth and the family's low socioeconomic status (SES) has been well documented. However, previous studies often focus on infants or preschoolers and primarily use static measures of vocabulary at multiple time points. To date, there is no research investigating whether SES predicts a child's word learning abilities in grade school and if so, what mediates this relationship. In this study, 68 children ages 8-15 years performed a written word learning from context task, which required using the surrounding text to identify the meaning of an unknown word. Results revealed that vocabulary knowledge significantly mediated the relationship between SES (as measured by maternal education) and word learning. This was true despite the fact that the words in the linguistic context surrounding the target word are typically acquired well before age 8. When controlling for vocabulary, word learning from written context was not predicted by differences in reading comprehension, decoding, or working memory. These findings reveal that differences in vocabulary growth between grade school children from low and higher SES homes are likely related to differences in the process of word learning more than knowledge of surrounding words or reading skills. Specifically, children from lower SES are not as effective at using known vocabulary to build a robust semantic representation of incoming text to identify the meaning of an unknown word.

Key words: Word learning, Socioeconomic Status, vocabulary, development

## **Introduction**

In the US, over 31 million children (42%) are from low socioeconomic status (SES) homes (U.S. Census, 2010). On average these children have significantly smaller vocabularies than their higher SES peers (Hart & Risley, 1995; Hoff, 2006). This vocabulary gap increases during the school years (Bradley & Corwyn, 2002) and has long-term academic implications (Hemphill & Tivnan, 2008). There is clear evidence that in infancy and preschool the primary reason for SES differences in the rate of vocabulary growth relates to differences in the quantity and quality of language exposure in the home (Hoff, 2006). The reason why the vocabulary gap increases throughout schooling is less well understood or researched. A primary question is whether early environmental differences related to SES lead to differences in the process of word learning in older children. Specifically, when grade schoolers are given the same word learning task, do children from low SES homes learn fewer words than those from higher SES homes and, if so, why? Given that word learning is foundational to vocabulary growth, identifying and explaining word learning problems in low SES children will provide new insights about why the vocabulary gap continues throughout grade school and thus potentially illuminate new avenues for intervention.

The process of word learning in grade school is highly dependent on learning from context- or deriving the word's meaning via exposure in multiple contexts (Bersten, et al., 2010; Nagy & Anderson, 1984). This type of word learning is an incremental process that includes identifying a new word, holding potential word meanings in memory, and eliminating incorrect meanings as new information becomes available (Fukkink, et al., 2001; Nagy, Herman, & Anderson, 1985). As would be expected of such a complex process, children vary greatly in this type of learning throughout schooling (Cain, Oakhill, & Elbro, 2003). Surprisingly, to our

knowledge, no research has directly studied how SES impacts learning from context in grade school. There is, however, a substantial literature identifying vocabulary knowledge, reading skills, and working memory as three influential processes on learning from context. Importantly, each of these has been identified as problematic for children from low SES homes.

One of the most consistent predictors of vocabulary growth in grade school and the most consistently reported area of language difficulty for children from low SES homes is vocabulary. Research unrelated to SES has shown that children with smaller vocabularies learn fewer words from context than children with larger vocabularies (Cain et al., 2003; Ricketts et al., 2011), even when the children know the meanings of all of the surrounding words (Daalen-Kaptein, et al, 2001; Swanborn & De Glopper, 1999, 2002). Thus, differences in word learning for school-aged children appear to be due to differences in the *process of learning*, as opposed to the knowledge of the individual words in the learning context.

Vocabulary knowledge is a particularly well documented area of difficulty for children from low SES homes (Weisleder & Fernald, 2013; Hoff, 2003, 2013). School-aged children from low SES homes make smaller gains on standardized vocabulary measures compared to children from higher SES homes (Farah, et al., 2006), resulting in an increasing vocabulary gap between the two groups. Surprisingly, while many theories attribute the continuing vocabulary gap to differences in vocabulary knowledge (Hemphill & Tivnan, 2008; Hoff, 2013; Christ & Wang, 2011), there do not appear to be any studies directly examining whether SES-related differences in vocabulary knowledge influence the process of word learning in grade school. This study addresses that question.

Reading skills, specifically decoding and comprehension, correlate with vocabulary knowledge (Carlson et al., 2013; Perfetti, McKeown, & Kucan, 2010); however, the literature is

divided regarding whether reading skills associate with word learning from context success in children. Some studies report an association (Swanborn & De Glopper, 2002; Steele & Watkins, 2010) and others do not (Nagy, Herman, & Anderson, 1985; Steele & Watkins, 2010). Like vocabulary knowledge, children from low SES homes have more reading difficulties than their higher SES peers (National Center for Education Statistics, 1999; Hackman & Farah, 2009). Disentangling the complex relationship between reading, vocabulary, SES, and word learning would help provide greater insights into language interventions in lower SES populations.

Outside of language-related skills, working memory is also correlated with word learning from context in adults and school-aged children (Cain, Oakhill, & Lemmon, 2004). Word learning from context requires one to hold in mind and actively integrate semantic information from multiple contexts surrounding the novel word, directly drawing on working memory. Children and adults from low SES homes often exhibit poorer working memory compared to higher SES peers (Farah, et al., 2006; Evans & Fuller-Rowell, 2013), which may influence how they learn words from context.

Here we investigate to what degree SES-related differences in word learning from linguistic context are mediated by vocabulary, reading abilities, and working memory. We use maternal education as our primary measure of SES because it has been the SES variable most strongly linked to language abilities (Hoff, 2013).

## **Method**

**Participants.** Eighty-two children between the ages of 8 and 15 participated. Fourteen were removed for parent's failure to report race (6), ethnicity (1), maternal education (3) or failure to complete all tasks (4), leaving 68 participants in the final analyses. Demographic information was obtained via parental questionnaire. Eighteen participants were bilingual, four

of whom learned Spanish and English simultaneously. In the bilingual population, the average age of English exposure was 3.4 years ( $SD=3.2$ ). SES was based on self-reported maternal education and grouped into 5 categories: Did not complete High School, High School Degree, Partial College, College Degree, and Graduate Degree. Appendix A provides demographic information.

**Procedure.** These data are the behavioral data from a larger EEG study (REMOVED TO DEIDENTIFY, in press). Participants came to our lab with their parents. After completing assent and consent, participants completed a battery of standardized tests and our experimental word learning from context task, with the order of the behavioral assessment and experimental tasks counterbalanced across participants. Vocabulary knowledge was tested using the Peabody Picture Vocabulary Test-4<sup>th</sup> Edition (PPVT-4; Dunn & Dunn, 2007). Reading was measured using the Test of Word Reading Efficiency-2<sup>nd</sup> Edition (TOWRE-2; Torgensen, Wagner, & Rashotte, 2012) and the Gray Oral Reading Tests-5<sup>th</sup> Edition (GORT-5; Wiederholt & Bryant, 2012). Specifically, we were interested in decoding using the TOWRE-2 phonemic decoding efficiency (PDE) measure and reading comprehension using the GORT-5. Working memory was measured using a digit span task. When available, scores standardized for age were used in the analysis; however some measures are not standardized. To adjust the scores for these measures, we accounted for age statistically in the model. All tasks were chosen based on their association with word learning in previous investigations of SES.

### **Word Learning From Context Task**

**Design.** Children read 100 sets of sentence triplets. Due to the nature of the EEG task design, the sentences were presented word-by-word on a computer screen. The last word of each sentence, referred to as the target word, was a novel word that the participant was to learn (see

Appendix B). There were 2 conditions: Meaning Plus (M+) and Meaning Minus (M-). In the M+ condition, sentences within each triplet increasingly supported the target word's meaning so that the last sentence provided a great deal of contextual support. The M- condition served as a control for repeated exposure to the target novel word without support for learning the word's meaning in the EEG task and thus unrelated to the primary questions in this paper. See REMOVED TO DEIDENTIFY (in press) for more detail on how the stimuli were created and tested. After each sentence triplet, children were asked to verbally report the word's meaning, if possible or respond that no word fit all sentences (M-). Responses were recorded and rated as correct or incorrect by two trained coders (interrater reliability was 96.8%). A correct response for M+ required participants' to provide the target word or a word that was rated as semantically plausible for three sentences, and an incorrect response was either no word or any other word. A correct response for M- was participants' response that no real word completed the triplet, and an incorrect response was any word. Participants completed a short training session before the task, including two M+ and two M- triplets during which they received accuracy feedback. No feedback was provided during the study.

**Stimuli.** Sentences were between 6 and 9 words in length. All of the words in the sentence were early-acquired, high frequency words from well-established corpora (Fenson et al., 1994; Hall, Nagy & Linn, 1984; Carroll, Davies, & Richman, 1971) thus removing knowledge of the specific words in the sentence as a confound in studying the word learning process. As opposed to teaching real word meanings that the children were unlikely to know, the sentences were created with a real word as the target word. Targets were concrete nouns in an average child's productive vocabulary by 30 months of age (Fenson et al., 1994) that were replaced with phonologically possible, mono-syllabic consonant-vowel-consonant novel words

(Storkel, 2013). Given the large differences in vocabulary across SES and age (8-15 year olds) we felt this was the best way to ensure that none of the children knew the words prior to the study. Further, children often learn new words with nuanced meanings for concepts they already have names for (Waxman & Senghas, 1992), anchoring the novel word's meaning to a known word, then identifying differences between the known and novel words' meanings with exposures. This project captured that early stage in the word learning process.

### Results

The goal of this study was to identify if there are differences in word learning related to SES, and if so, if they could be explained by vocabulary, reading skills, and/or working memory. We used the PROCESS macro (Hayes, 2013) to specify this parallel multiple mediator model with ordinary least squares path analysis (see Figure 1). To ensure that demographic features (i.e., age, ethnicity, race, sex, and bilingualism) did not provide an alternative explanation for the effects of SES (operationalized as maternal education) on the outcomes, we controlled for these variables in the multiple mediation analysis.<sup>1</sup> See Table 1 for the partial correlations when controlling for age, sex, bilingualism, ethnicity and race. Figure 1 shows that better educated mothers had children with more advanced vocabularies. Moreover, children with more advanced vocabularies displayed a greater percentage of words learned. No other relations emerged as significant in the model.

To further evaluate whether vocabulary knowledge, decoding, reading comprehension, and/or working memory mediated the relation between maternal education and percentage of words learned, we computed indirect effects for the specific pathways (labeled “ab” in Figure 1) and used bias-corrected bootstrapping with 5,000 samples to construct 95% confidence intervals

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<sup>1</sup> To partial out the effects of demographic features, control variables were included as predictors in the analyses regressing each of the mediators on Maternal Education, as well as the analysis regressing percentage of words learned on the mediators and Maternal Education.



for the indirect effects (intervals not containing zero indicate that the indirect effect is statistically significant)<sup>2</sup>. We also computed completely standardized indirect effects (labeled “abcs” in Figure 1) to obtain measures of effect size (Preacher & Hayes, 2008); values of  $|.01|$ ,  $|.09|$ , and  $|.25|$  are considered small, medium, and large effects, respectively.

Only the Maternal Education-to-Vocabulary Knowledge-to-Percentage of Words Learned pathway emerged as significant (although this indirect effect was significantly larger than the Maternal Education-to-Reading Comprehension-to-Percentage of Words Learned pathway [difference = 0.03, 95% CI [0.01, 0.07]], it did not significantly differ from any of the other pathways). Further, the completely standardized indirect effect for this pathway was medium-to-large in size, suggesting that vocabulary knowledge represents an important mechanism linking maternal education to the percentage of words learned.

Because bilingualism may interact with other variables in the analysis, we also treated bilingualism as a moderator of the Maternal Education-to-mediator paths, the mediator-to-Percentage of Words Learned paths, the direct effect for Maternal Education, and the four indirect effects. Bilingualism was not a moderator of the direct effect of SES on word learning or on any of the indirect pathways between SES and word learning. Bilingualism only significantly moderated the effect of decoding on percentage of words learned,  $b = 0.004$ ,  $SE = .002$ ,  $p = .034$ . For bilingual children, higher decoding scores were significantly related to a greater percentage of words learned ( $b = 0.003$ ,  $SE = .002$ ,  $p = .049$ ); decoding was unrelated to percentage of words learned in monolingual children ( $b = -0.005$ ,  $SE = .003$ ,  $p = .13$ ).

## Discussion

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<sup>2</sup> Bootstrapping refers to randomly sampling the data with replacement. In bias-corrected bootstrapping, the resulting percentiles used to construct the lower limits and upper limits of the confidence interval are corrected for the possible difference between the observed indirect effect and the mean indirect effect from the bootstrapped sampling distribution.

The goal of this project was to identify if SES was related to word learning in grade school and to what degree vocabulary, reading and working memory might mediate that relationship. We found that the relationship between SES and word learning was significantly mediated by vocabulary, but not other processes known to be related to word learning and problematic for children from low SES homes: reading (both comprehension and decoding) and working memory. These findings provide new insights into why the vocabulary gap between low and higher SES children continues to grow throughout grade school and what strategies may be effective for interventions.

One of the most important contributions of this study is the focus on word learning as opposed to a static measure of vocabulary growth. These findings allow us to discuss issues related to process, as opposed to predictors. It is well-documented that children from lower SES homes exhibit lower vocabulary scores and less growth in vocabulary scores over time than their higher SES peers. However, work showing that effect has not clarified whether the discrepancy is primarily driven by differences in the linguistic input, as is reported in younger children, or differences in the word learning process. Our study shows that skills related to the process of word learning may be one reason for the continued discrepancy in vocabulary growth in grade school. Our findings do not rule out differences in the linguistic environment, which are likely due to differences in the home and differences between schools in low and high income areas. However, by pinpointing word learning differences as problematic we can provide better language interventions and instruction to children from low SES homes.

Our findings indicate that providing strategies for children to better utilize surrounding linguistic context when learning a new word, such as trying to form pictures of the context they are reading as they read, may be more beneficial for vocabulary growth than just increasing a

child's vocabulary size. We draw this conclusion based on the fact that the words used in this task were well within the children's comprehension and production abilities. The reason that vocabulary played such a strong role in word learning is likely attributed to the relationship between vocabulary size and depth of word knowledge (Cain et al., 2003; Weisleder & Fernald, 2013). When children have a deeper knowledge of words in context, they can integrate new semantic information into a deeper web of information to better predict what will come next. When encountering an unfamiliar word, this predictive information helps in meaning identification. We suspect that children from low SES homes have difficulties consolidating information related to known words to build a robust semantic representation that helps them determine an unknown word's meaning.

In the current study, bilingual children were overrepresented in lower SES groups. Our mediation analysis accounted for this statistically, but there may be differences related to bilingualism that were not identified. For example, children who are bilingual use mutual exclusivity somewhat differently than their monolingual peers (Davidson & Tell, 2005). As such the two populations may be performing this task differently. One way we tried to address the issue of bilingualism was to treat it as a moderator in our model. Although bilingualism moderated the effect of phonological decoding on word learning, the results revealed no moderating effect of bilingualism on the direct and indirect relationships between SES and word learning. Thus, the importance of vocabulary as a mediating factor on the relationship between SES and word learning held true regardless of language status. However, this result highlights the complexity of the relationship between SES, bilingualism, and word learning and the need for more research in this area.

The finding that reading comprehension and decoding were not significant mediating factors in the relationship between maternal education and word learning is particularly interesting. This is likely because the reading demands in this task were relatively low. As noted, the words in the written context were high frequency, early-acquired words, making decoding relatively simple. Further, the sentences in each triplet were unrelated to one another; thus, children did not need to follow a storyline to determine the meaning of the word. More difficult words in a continuing narrative may tax reading skills such that their influence on word learning would be more significant. This is an important avenue for future investigation, however, by alleviating the level of reading skill needed to complete the current task, we believe the current findings reflect important differences in general word learning that are not confined to learning from written text.

One limitation of our study is that designing it for EEG required words to be presented individually likely increasing the working memory load, but we found no correlation between working memory and maternal education nor was working memory a significant predictor in our model. Therefore, we do not believe that working memory demands explain our findings.

Despite increased attention to the importance of vocabulary development following Hart and Risley's (1995) groundbreaking work over 20 years ago, the vocabulary gap between low and high SES children persists. In this paper, we took a unique approach to study the process of word learning as opposed to predictors of vocabulary growth. Our findings reveal important new information about how and why vocabulary knowledge impacts word learning and provides new insights about potential interventions or teaching techniques for low SES populations.

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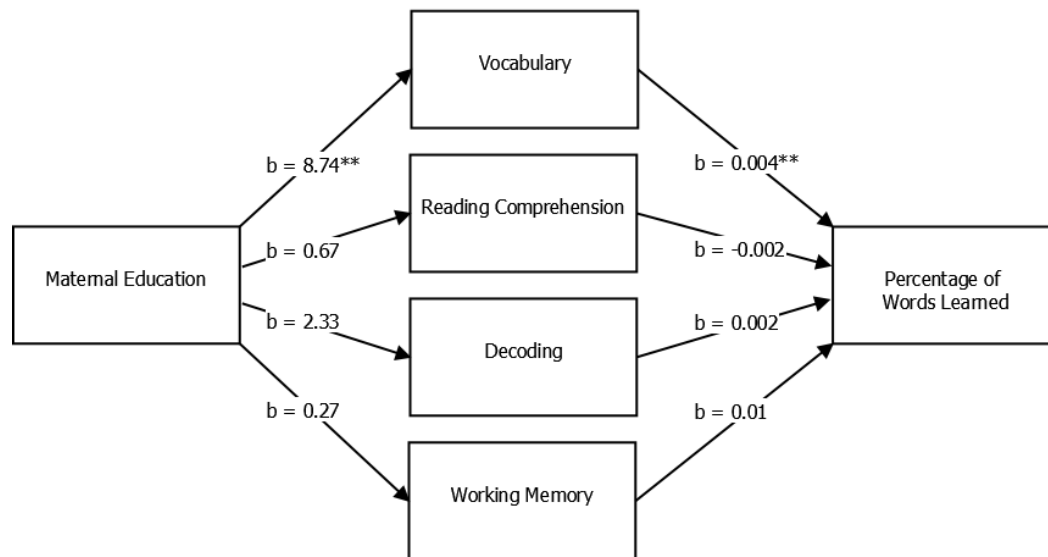
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Figure 1. Multiple Mediation Model (controlling for age, sex, race, ethnicity, and bilingualism)



ME -> Vocab -> % of Words Learned:  $ab = 0.03 [0.01, 0.07]$ ;  $abcs = 0.19 [0.06, 0.38]$

ME -> Read Comp -> % of Words Learned:  $ab = -0.00 [-0.02, 0.00]$ ;  $abcs = -0.01 [-0.12, 0.02]$

ME -> Decoding -> % of Words Learned:  $ab = 0.00 [-0.00, 0.03]$ ;  $abcs = 0.03 [-0.02, 0.14]$

ME -> WM -> % of Words Learned:  $ab = 0.00 [-0.00, 0.02]$ ;  $abcs = 0.02 [-0.02, 0.13]$

Note: Paths in Figure represent unstandardized regression coefficients. ME = Maternal Education.

Vocab = Vocabulary. Read Comp = Reading Comprehension. WM = Working Memory.

$ab$  = unstandardized indirect effect.  $abcs$  = completely standardized indirect effect.

95% Confidence Intervals for the indirect effects are contained in brackets after the point estimates, and were constructed using bias-corrected bootstrapping with 5,000 samples.  $^{**}p < .006$ .

Table 1. Partial Correlations between Study Variables (Controlling for Age, Sex, Bilingualism, Ethnicity, and Race

| Variable                  | 1     | 2     | 3     | 4     | 5     | 6 |
|---------------------------|-------|-------|-------|-------|-------|---|
| 1. Maternal Education     | -     |       |       |       |       |   |
| 2. Perc. of Words Learned | .16   | -     |       |       |       |   |
| 3. Vocabulary             | .47** | .47** | -     |       |       |   |
| 4. Reading Comprehension  | .12   | .16   | .29*  | -     |       |   |
| 5. Decoding               | .15   | .33*  | .29*  | .33** | -     |   |
| 6. Working Memory         | .10   | .37** | .37** | .38** | .42** | - |